



Aror University of Art, Architecture, Design & Heritage Sukkur

Department of Artificial Intelligence and Multimedia Gaming

CS-207 Database Systems (BS-AI)

LAB # 02

Lab Manual: Exploring Gallery and Creating Various Forms Reports in Oracle APEX

Objective:

To familiarize students with Oracle APEX's Gallery and its applications, and to guide them in creating Faceted Reports, Master-Detail Reports, and other report types.

Lab Activity 1: Exploring Oracle APEX Gallery

Purpose: Understand the Oracle APEX App Gallery and its features to accelerate application development.

Steps:

1. **Login to Oracle APEX:**
 - o Open Oracle APEX and log in using your workspace credentials.
 - o Navigate to the "App Builder" section.
 2. **Explore the App Gallery:**
 - o Click on **Gallery** under the "App Builder" menu.
 - o Browse through sample applications like Sample Database Application, Sample Charts, or Sample Interactive Grid.
 - o Select one application, open its details, and run it to understand its functionalities.
 3. **Analyze Features:**
 - o Note the pre-built reports, dashboards, and forms in the sample app.
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Lab Activity 2: Creating a Faceted Report

Purpose: Learn to design dynamic faceted reports for effective data filtering.

Steps:

1. **Setup:**
 - o From the App Builder, create a new application.
 - o Select a data source (e.g., a table from your database).
 2. **Add a Faceted Search Page:**
 - o While creating the application, choose "Faceted Search" as one of the pages.
 - o Configure the facets (e.g., filtering by category, region, or status).
 3. **Customize Facets:**
 - o Go to the "Page Designer."
 - o Modify the facets by adjusting filters (e.g., slider for numeric values or dropdown for categories).
 4. **Run and Test:**
 - o Run the application.
 - o Test the functionality by filtering data using different facets.
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Lab Activity 3: Creating a Master-Detail Report

Purpose: Build a report with a parent-child relationship to display detailed data based on master data selection.



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Steps:

1. **Setup Master Table:**
 - o Use two tables with a parent-child relationship (e.g., Orders and Order Details).
 2. **Add Master-Detail Pages:**
 - o In the App Builder, create a new page.
 - o Select "Master-Detail" as the page type.
 - o Configure the relationship between the two tables.
 3. **Run and Test:**
 - o Run the application and navigate through the Master-Detail report.
 - o Verify the child data updates based on the selected master row.
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Lab Activity 4: Creating Other Report Types

Purpose: Explore other types of reports like Interactive Reports and Classic Reports.

Steps:

1. **Interactive Report:**
 - o Create a new page and select "Interactive Report."
 - o Choose a table or SQL query as the data source.
 - o Customize filters, formatting, and actions.
 2. **Classic Report:**
 - o Add a "Classic Report" page.
 - o Use the Page Designer to modify its appearance (e.g., headers, pagination).
 3. **Charts and Dashboards:**
 - o Create a new page and select "Chart."
 - o Choose a chart type (e.g., Bar, Line, Pie).
 - o Configure the data series and labels.
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Lab Submission Tasks: Create a submission file and mention steps.

1. Create two Apps using Customer Order and HR datasets.
2. Explore Sample as well as Starter Apps.
3. Create a faceted report for filtering employee data by department, job title, and salary range.
4. Develop a Master-Detail report for displaying customer orders and order details.
5. Create Interactive grid and Classic reports and describe the difference between them.
6. Add an interactive dashboard with charts summarizing sales by region.